

Exploratory Data Analysis (EDA) on Titanic Dataset

Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to identify patterns, trends, relationships, and factors affecting passenger survival.

Dataset Information

Dataset Name: Titanic Dataset (train.csv)

Number of Records: 891

Number of Features: 12

Tools Used:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

Analysis Performed

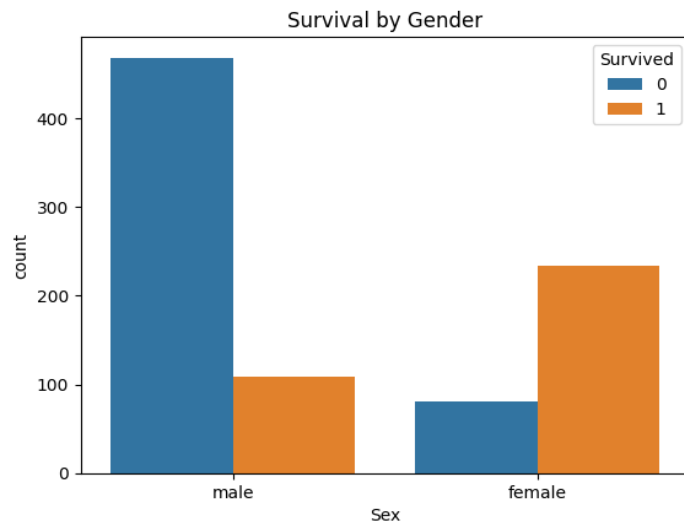
1. Dataset Overview

The dataset was examined using shape, columns, info, and describe functions to understand its structure and statistical properties.

2. Missing Value Analysis

Missing values were identified in the Age, Cabin, and Embarked columns.

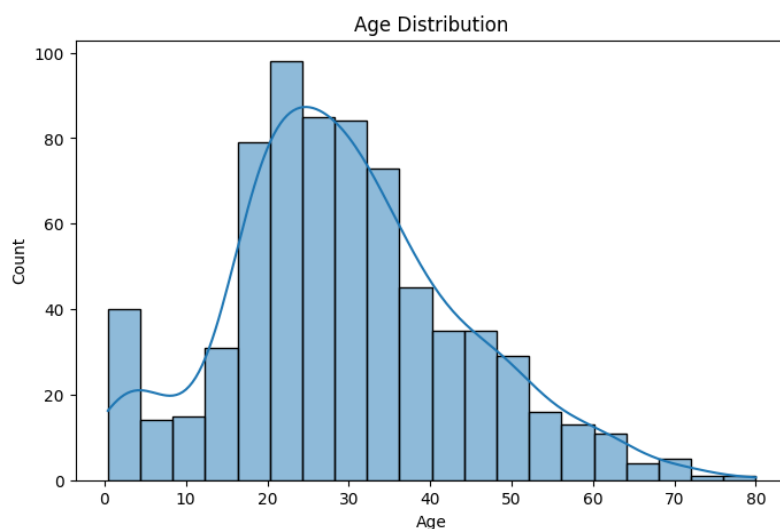
3. Survival Analysis by Gender



Observation:

Female passengers had a significantly higher survival rate than male passengers.

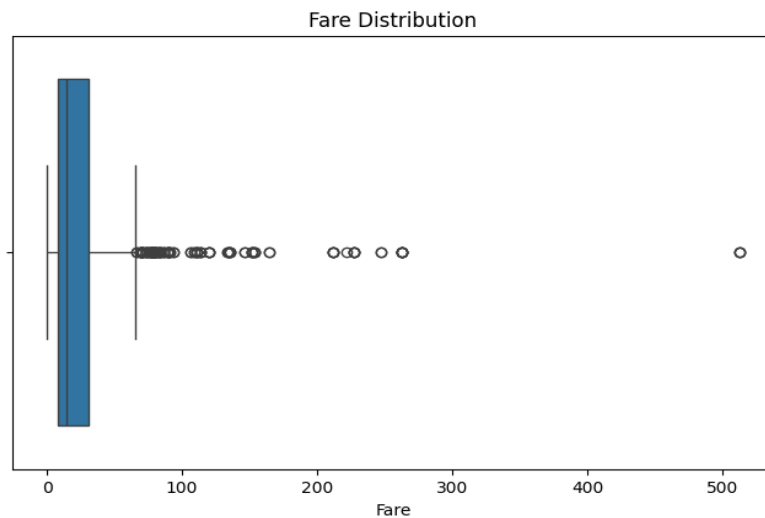
4. Age Distribution



Observation:

Most passengers were between 20 and 40 years old.

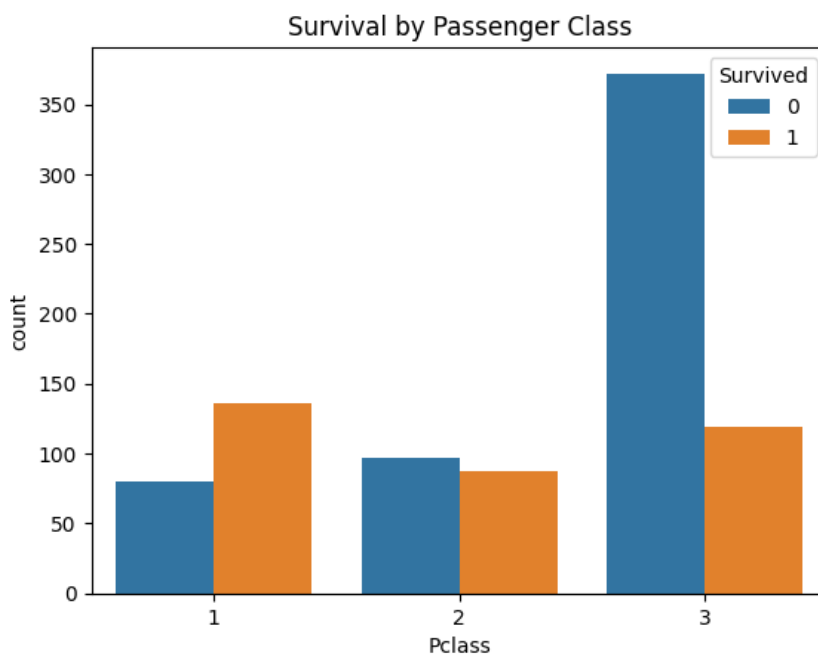
5. Fare Distribution



Observation:

Several outliers were present in fare values, indicating a wide variation in ticket prices.

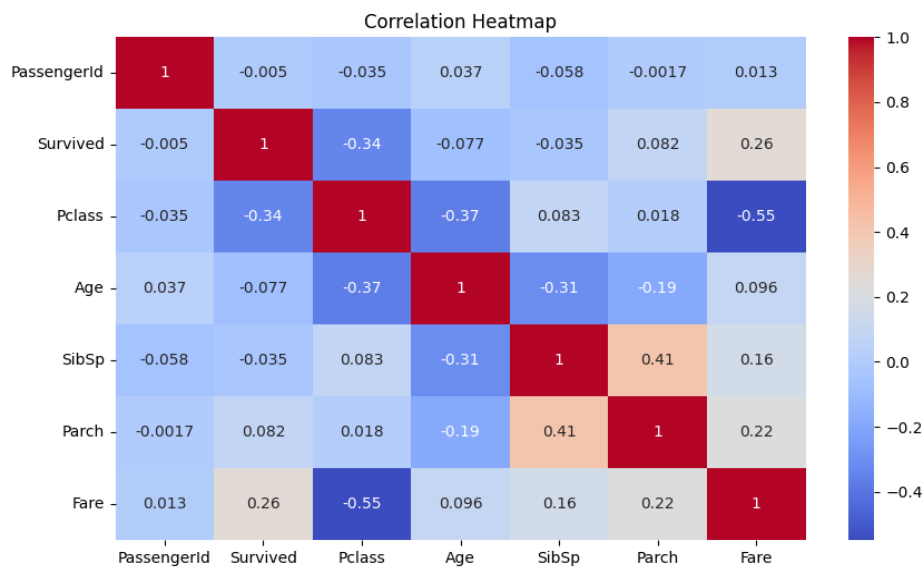
6. Survival by Passenger Class



Observation:

First-class passengers had better survival chances compared to lower classes.

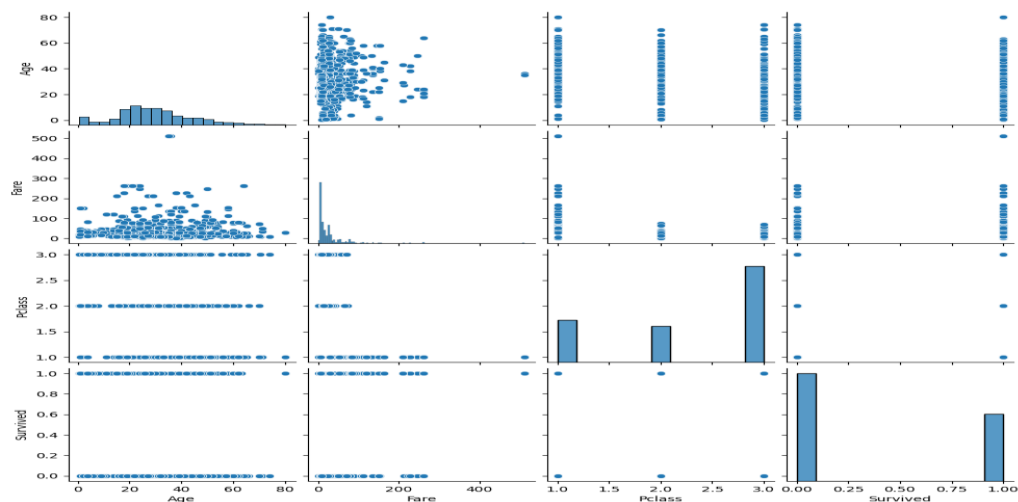
7. Correlation Analysis



Observation:

Passenger class and fare showed relationships with survival.

8. Additional Analysis



Observation:

The pairplot shows relationships among Age, Fare, Passenger Class, and Survival. Passengers paying higher fares generally belonged to higher classes and had better survival chances

Key Findings

1. More passengers died than survived.
2. Female passengers had a significantly higher survival rate than males.
3. Most passengers were between 20 and 40 years old.
4. First-class passengers had better survival chances.
5. Fare values contained several outliers.
6. Passenger class and fare influenced survival.
7. Missing values were mainly present in the Age and Cabin columns.
8. Survival rates varied across age groups and embarkation ports.
9. Gender, passenger class, and fare were the most influential factors affecting survival.

Conclusion

The Exploratory Data Analysis revealed important patterns and insights from the Titanic dataset. Gender, passenger class, and fare were identified as the most significant factors affecting passenger survival. The analysis demonstrates the importance of EDA in understanding data and discovering meaningful relationships before applying predictive models.